Finite-width circuit learning below mean-field scaling

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Abstract

For fixed-depth Bayesian tanh networks with Gaussian scaling exponent $\gamma < 1$, we prove a finite-width sample bound of order $\mathrm{poly}(d, C)N\log(eN)$, where d is input dimension, C is the target's bounded-weight network complexity, and $N \geq C$ is the hidden width. The prior is restricted to a fixed positive output margin on the entire Boolean cube, followed by exact noiseless conditioning on the training labels. The guarantee concerns the actual posterior: its correct-label probability exceeds $2/3$ on almost all test inputs. Consequently, polynomial widths permit polynomial-sample learning of parity, and every subexponential width sequence permits subexponential sample complexity. These statements concern simultaneous growth of width and dimension. They do not settle sample complexity when width first tends to infinity at each fixed dimension. The proof gives an explicit Gaussian probability bound by cloning a small target network and constraining only linearly many parameter directions in width.

1 Model, theorem, and order of limits

Let $d \geq 1$, $L \geq 2$, and $\mathcal{X} = \{-1, 1\}^d$. A depth-$L$ network with common hidden width N computes

$$h^0(x) = (1, x), \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \quad (1 \leq s < L), \qquad z_w(x) = W^L h^{L-1}(x).$$

There are no additional biases, and the readout is linear. Fix a real exponent $\gamma < 1$. Under the product Gaussian law $\mu_{N,\gamma}$, the independent centered weights have standard deviations

$$\sigma_1 = N^{(1-\gamma)/2}/\sqrt{d}, \qquad \sigma_s = N^{-\gamma/2} \quad (2 \leq s < L), \qquad \sigma_L = N^{-(\gamma+1)/2}. \tag{1}$$

Fix an output margin $a > 0$. For each Boolean function g, write

$$A^a_{N,g} = \{w : g(x)z_w(x) > a \text{ for every } x \in \mathcal{X}\}, \qquad \pi_N(g) = \frac{\mu_{N,\gamma}(A^a_{N,g})}{\mu_{N,\gamma}(\bigcup_h A^a_{N,h})}.$$

Thus the global margin restriction precedes training. The posterior $\Pi_N(\cdot \mid S_n)$ conditions this function prior on agreement with noiseless examples $S_n = ((x_i, f(x_i)))_{i=1}^n$, with independent $x_i \sim D$. The distribution D may be arbitrary on the cube.

Let $C = C_L(f)$ be the least integer $C \geq 1$ for which a depth-L, width-C tanh network implements f with strict margin $1/2$ and all weights bounded in absolute value by C. We assume $C < \infty$. The

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definition uses margin $1/2$ even when the Bayesian margin a differs; a constant readout rescaling handles this distinction. Define

$$R_N(S_n) = \mathbb{E}_{g\sim\Pi_N}\mathbb{P}_{x\sim D}\{g(x) \neq f(x)\},$$
$$F_N(S_n) = \mathbb{P}_{x\sim D}\left\{\Pi_N(g(x) = f(x) \mid S_n) \leq 2/3\right\}.$$

The equality case at $2/3$ is counted as a failure.

Here are explicit constants for the main bound:

$$b_L = \sqrt{16\log 9 + 8\log(4L)}, \qquad A = 2 + 3C^2 + b_L,$$
$$D_0 = 1 + 4a\sqrt{2}\,C^2(d+2)A^{L-2}, \qquad \rho_0 = \frac{a}{2D_0},$$
$$q = d + 2 + (L-2)C, \qquad B_0 = d(d+1)C^2 + 2(L-2)C^4 + 64a^2C^4. \tag{2}$$

For every $N \geq C$, set

$$B_N = B_0 N^\gamma + Nq\left[\log\frac{e\sqrt{2\pi}}{2\rho_0} + \frac{(L-1)(1-\gamma)}{2}\log N\right] + \log 2. \tag{3}$$

Theorem 1.1 (Finite-width learning)**.** *For every $N \geq C$, the target has raw prior mass*

$$\mu_{N,\gamma}(A^a_{N,f}) \geq e^{-B_N}. \tag{4}$$

For every input distribution D and every $\varepsilon, \delta \in (0,1)$, if

$$n \geq \frac{6}{\varepsilon}\left[B_N + \log\frac{6}{\varepsilon\delta}\right], \tag{5}$$

then, with probability at least $1-\delta$ over S_n,

$$R_N(S_n) \leq \varepsilon/3, \qquad F_N(S_n) \leq \varepsilon.$$

At fixed L, γ, a, the cost obeys

$$B_N \leq p_{L,\gamma,a}(d, C)\, N\log(eN) \tag{6}$$

for a polynomial $p_{L,\gamma,a}$ independent of width.

Corollary 1.2 (Polynomial and subexponential widths)**.** *At fixed $L, \gamma, a, \varepsilon, \delta$, widths bounded by a polynomial in d, $C_L(f)$ admit polynomial sample complexity under the actual posterior. For full parity on k bits, $C_L(f) = O_L(k\log(k+1))$. Consequently, for every integer width sequence $N(k) \geq C_L(f)$ satisfying $\log N(k) = o(k)$, there is a sample schedule $n(k) = \exp(o(k))$ for which $F_{N(k)}(S_{n(k)}) \leq \varepsilon$ with probability at least $1-\delta$. In particular, polynomial widths admit polynomial-sample parity learning.*

The order of limits is part of the conclusion. The corollary permits width to grow with dimension, for example $N(k) = \lceil e^{\sqrt{k}}\rceil$. A different question fixes k and its sample budget and then sends $N \to \infty$. The sample bound (5) grows with N, so it does not establish an upper bound for that width-first limiting learner. Nor does it contradict a lower bound proved for that learner. An exponential-in-k obstruction asserted uniformly over growing widths, however, would conflict with Corollary 1.2.

The guarantee is statistical. No efficient method for evaluating the posterior is asserted. The following proofs are finite-dimensional and do not require a Gaussian-process approximation.

2 Cloning the target at an arbitrary width

Choose a width-C implementation of f from its complexity definition. Multiply its readout by $4a$. Write its hidden matrices as B^s and readout as v. The resulting teacher has strict margin greater than $2a$, hidden entries bounded by C, and readout entries bounded by $4aC$. Partition the N neurons into C groups with sizes k_b differing by at most one. Since $N \geq C$,

$$k_b \geq 1, \qquad \frac{\max_b k_b}{\min_b k_b} \leq 2, \qquad \frac{N}{k_b} \leq 2C. \tag{7}$$

The exact clone w_* has first-layer row B^1_b in group b, internal entry B^s_{ba}/k_a from group a to group b, and readout entry v_b/k_b in group b. It computes precisely the teacher's output. Each hidden activation is constant within every group.

Let $u_a = k_a^{-1/2}\mathbf{1}_{\text{group } a}$ and let P be the orthogonal projection onto their span. Restricting an internal cloned matrix to this subspace gives entries $\sqrt{k_b/k_a}B^s_{ba}$. Its Frobenius norm and (7) imply

$$\|W^s_*\|_{\text{op}} \leq \sqrt{2}\,C^2 \quad (2 \leq s < L), \qquad \sqrt{N}\|W^L_*\|_2 \leq 4a\sqrt{2}\,C^2. \tag{8}$$

The cloned internal matrices vanish on the orthogonal complement of the same group-constant subspace.

Put

$$t = N^{(1-\gamma)/2} \geq 1, \qquad \rho = \rho_0 t^{-(L-1)}. \tag{9}$$

The normalized coordinates $\sqrt{d}\,W^1_{ij}$, $\sqrt{N}\,W^s_{ij}$ for $2 \leq s < L$, and $\alpha_j = NW^L_j$ are independent centered Gaussians with standard deviation t. Impose the following cylinder event $\mathcal{C}_\rho$:

- Every $\sqrt{d}\,W^1_{ij}$ is within ρt of its cloned value.
- For each internal row j and group a, the coordinate $\langle\sqrt{N}\,W^s_{j,\cdot}, u_a\rangle$ is within ρt of its cloned value $\sqrt{N/k_a}B^s_{ba}$, where j belongs to group b.
- Every α_j is within ρt of its cloned value Nv_b/k_b.

These are Nq constraints on independent scalar Gaussian coordinates. The orthogonal coordinates of each internal row retain their Gaussian law and remain independent of the constrained coordinates. Thus

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^sP, \quad E^sP = 0, \qquad \|R^s\|_{\text{op}} \leq \rho t\sqrt{C}.$$

Conditionally on the cylinder, each free matrix E^s has law $(t/\sqrt{N})G^s(I-P)$ with G^s a standard Gaussian matrix.

Lemma 2.1 (Free Gaussian directions)**.** *Conditionally on $\mathcal{C}_\rho$, all internal free matrices satisfy $\|E^s\|_{\text{op}} \leq b_L t$ with probability at least $1/2$.*

Proof. A $1/4$-net of the unit sphere in $\mathbb{R}^N$ has at most 9^N points, by the disjoint-ball volume argument. Approximating both vectors in a bilinear form gives $\|G\|_{\text{op}} \leq 2\max_{u,v \text{ in the net}} |u^TGv|$. Each such bilinear form for standard Gaussian G is standard normal. The scalar Gaussian tail bound gives

$$\mathbb{P}\{\|G\|_{\text{op}} > b_L\sqrt{N}\} \leq 2\,9^{2N}e^{-b_L^2N/8} = 2(4L)^{-N} \leq \frac{1}{2L}.$$

Use $\|G(I-P)\|_{\text{op}} \leq \|G\|_{\text{op}}$ and a union bound over the $L-2$ internal matrices. For $L=2$ the event is automatic. □

Lemma 2.2 (Uniform preservation of the target margin)**.** *Every draw in $\mathcal{C}_\rho$ satisfying the event of Lemma 2.1 implements f with strict margin greater than $3a/2$ on the entire cube.*

Proof. The choice of D_0 implies $0 < \rho_0 < 1$, and therefore $\rho t = \rho_0 t^{-(L-2)} \le 1$. For each input define $e_s = N^{-1/2}\|h_w^s - h_*^s\|_2$. The first-layer constraints and tanh's Lipschitz constant give $e_1 \le \rho t(d+1)$. Since $E^s h_*^{s-1} = 0$, unit boundedness of the hidden activations and (8) give

$$e_s \le (\sqrt{2}C^2 + \rho t\sqrt{C} + b_L t)e_{s-1} + \rho t\sqrt{C} \le tAe_{s-1} + \rho t\sqrt{C}.$$

Summing the geometric series and using $\sqrt{C}/(tA-1) \le 1$ yields

$$e_s \le \rho t^s (d+2)A^{s-1}.$$

In particular, $e_{L-1} \le \rho_0(d+2)A^{L-2}$. At the output the perturbation is bounded by

$$|z_w - z_*| \le 4a\sqrt{2}\,C^2 e_{L-1} + \rho t \le \rho_0[4a\sqrt{2}\,C^2(d+2)A^{L-2} + 1] = a/2.$$

These bounds hold simultaneously for every input, without a union bound on cube points. The teacher has strict margin greater than $2a$. □

3 The probability cost and posterior bound

Lemma 3.1 (Cylinder probability)**.** *The target-mass bound (4) holds with B_N from (3).*

Proof. The sum of squared centers of the Nq constrained normalized coordinates is at most NB_0. A first-layer center has magnitude at most $\sqrt{d}C$. An internal center has magnitude at most $\sqrt{2C}\,C$. A readout center has magnitude at most $8aC^2$. Counting the coordinates of each kind gives the expression for B_0 in (2).

For a centered Gaussian Z of standard deviation t, integrating the minimum density on an interval and using $(|c|/t + \rho)^2/2 \le c^2/t^2 + \rho^2$ gives

$$\mathbb{P}\{|Z - c| \le \rho t\} \ge \frac{2\rho}{\sqrt{2\pi}} e^{-c^2/t^2 - \rho^2}.$$

Multiply over the independent constrained coordinates and use $\rho \le 1$. Lemmas 2.1 and 2.2 imply

$$\begin{aligned} -\log \mu_{N,\gamma}(A^a_{N,f}) &\le \frac{NB_0}{t^2} + Nq \log \frac{e\sqrt{2\pi}}{2\rho} + \log 2 \\ &= B_0 N^\gamma + Nq\left[\log \frac{e\sqrt{2\pi}}{2\rho_0} + \frac{(L-1)(1-\gamma)}{2}\log N\right] + \log 2. \end{aligned}$$

This is B_N. Because $N^\gamma \le N$, while B_0, q, and ρ_0^{-1} are polynomially bounded in d, C at fixed L, a, (6) follows. □

Lemma 3.2 (A target-mass posterior guarantee)**.** *Let π be a prior on a finite class of Boolean functions and suppose $\pi(f) \ge e^{-B} > 0$. Under exact consistency conditioning on n independent noiseless examples, the sample bound*

$$n \ge (6/\varepsilon)[B + \log(6/(\varepsilon\delta))]$$

implies $R \le \varepsilon/3$ and $F \le \varepsilon$ with probability at least $1 - \delta$.

Proof. Let $Z(S_n)$ be the posterior normalizer. The target is always consistent, so $Z(S_n) \geq \pi(f)$. A fixed function of true error r survives all labels with probability $(1-r)^n$. Thus, if $U(S_n)$ denotes posterior mass on functions of error greater than $\varepsilon/6$,

$$\mathbb{E}_{S_n} U(S_n) \leq \frac{1}{\pi(f)} \sum_{g:\mathrm{err}_D(g)>\varepsilon/6} \pi(g)(1-\mathrm{err}_D(g))^n \leq e^{B-n\varepsilon/6}.$$

Markov's inequality gives $U \leq \varepsilon/6$ with probability at least $1-\delta$ at the displayed sample size. The remaining posterior mass has error at most $\varepsilon/6$, so $R \leq \varepsilon/3$. On an input with correct-label probability at most $2/3$, the posterior error probability is at least $1/3$. Consequently $F \leq 3R \leq \varepsilon$, including equality in the failure event. □

Proof of Theorem 1.1. Lemma 3.1 proves the raw mass bound and its growth estimate. The global-margin normalizer is at most one, so $\pi_N(f) \geq \mu_{N,\gamma}(A^a_{N,f}) \geq e^{-B_N}$. In particular all posterior normalizers are positive. Apply Lemma 3.2. □

4 Parity and simultaneous growth

Lemma 4.1 (A small parity network)**.** *At every fixed depth $L \geq 2$, full parity on k bits has $C_L(f) = O_L(k \log(k+1))$.*

Proof. Put

$$t(x) = \frac{k + \sum_{i=1}^k x_i}{2}, \qquad y_j = (-1)^{k-j}, \qquad A_k = \log(16(k+1)).$$

At integer $t \in \{0, \ldots, k\}$, define $s_j(t) = \mathrm{sign}(t - j + 1/2)$ for $1 \leq j \leq k$. The identity

$$y_t = c + \sum_{j=1}^k y_j s_j(t), \qquad c = (y_0 + y_k)/2 \in \{0, 1\},$$

is checked by its value at $t = 0$ and its jumps. Replace s_j by $h_j = \tanh(A_k(t - j + 1/2))$. Each replacement has error at most $2e^{-A_k}$, since its argument has absolute value at least $A_k/2$. A constant neuron $h_0 = \tanh 1$ provides the term c with readout weight $c/\tanh 1$. The resulting output differs from parity by less than $1/8$. This network has width $k+1$, first-layer weights bounded by $O(k \log(k+1))$, and readout weights bounded by two.

For $L \geq 3$, let the next hidden layer apply tanh to this sum. Its correct-signed output is greater than $\tanh(7/8) > 1/2$. Every additional hidden layer preserves the margin using a single weight two, and the final readout has weight one. Pad unused neurons with zero weights. Finally pad the common width to an integer $C = O_L(k \log(k+1))$ that also bounds every weight. Since $\tanh 0 = 0$, this preserves the implementation. □

Proof of Corollary 1.2. The polynomial-width statement follows directly from (6) and (5). For parity use Lemma 4.1. At fixed accuracy and confidence parameters, the sufficient sample size is bounded by a polynomial in k times $N(k)\log(eN(k))$. Its logarithm is $o(k)$ whenever $\log N(k) = o(k)$. □

The cylinder uses $N[d + 2 + (L-2)C]$ constrained directions even though a dense deep network has quadratically many parameters in width. The other internal directions are retained at their prior scale; their effect is controlled by shrinking the constrained radius. For $\gamma < 1$, this shrinking costs an additional $\log N$ per constrained direction. That finite-width cost is why these learning guarantees coexist with possible statistical obstructions after taking width to infinity first.